OR-AS Operations Research Applications and Solutions	Case Name: **Premium Payment System**	Sector	**IT**	
	OR-AS Operations Research - Applications and Solutions www.or-as.be info@or-as.be	Baseline Schedule	Schedule with resources	
			Schedule with costs	
		Risk Analysis	Random simulation	
			One of nine std. scenarios	
Submitted by	**Matilde Casado and Margarida Antunes**		User defined distributions	
Date	**2015**	Project Control	Automatic tracking	
File Name	**C2015-12 Premium Payment System**		Tracking based on user input	

1. Project description

Project authenticity

An IT project involving analysis, design, development, testing, and implementation to enhance system processes and payment handling.

The project consists of activity, resource and cost data that were obtained directly from the actual project owner.

2. Project properties

2.1. Baseline Schedule

General	
# Activities	42
Planned Duration (PD)	184 days*
Budget At Completion (BAC)	132.570 €
Renewable Resources	17
Consumable Resources	-

* standard eight-hour working days

Network topology	
Serial/Parallel (SP)	19%
Activity Distribution (AD)	63%
Length of Arcs (LA)	9%
Topological Float (TF)	61%

2.2. Risk Analysis

Random simulation by ProTrack was performed using the default symmetric triangular risk distribution profiles.

	Cost sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	11.7	13.4	-2.2
CRI-rho	20.8	18.7	-1.0
CRI-tau	26.8	36.9	-1.5

	Resource sensitivity		
	avg [%]	std dev [%]	skew [-]
CRI-r	11.7	10.2	-3.9
CRI-rho	11.7	10.2	-3.9
CRI-tau	11.7	10.2	-3.9

	Time sensitivity		
	avg [%]	std dev [%]	skew [-]
CI	10.6	18.0	-1.6
SI	27.9	26.0	-0.8
SSI	4.8	13.2	-4.0
CRI-r	9.3	12.4	-2.8
CRI-rho	19.0	19.1	-1.0
CRI-tau	27.3	36.6	-1.5

2.3. Project Control

2.3.1. Simulated forecasting accuracy

The accuracy of time and cost forecasting methods has been evaluated based on Monte Carlo simulation runs using the risk profiles described in section "2.2. Risk Analysis". Based on these risk profiles, the Mean Absolute Percentage Error (MAPE) and Mean Percentage Error (MPE) has been calculated to evaluate the expected accuracy of the time and cost predictions, EAC(t) and EAC, respectively.

Simulated EAC(t) accuracy		
method - PF	MAPE [%]	MPE [%]
PV - 1	217.2	-217.2
PV - SPI	165.0	-143.5
PV - SCI	163.8	-141.2
ED - 1	310.2	-310.2
ED - SPI	165.0	-143.5
ED - SCI	163.9	-141.7
ES - 1	258.4	-258.4
ES - SPI(t)	67.6	-67.6
ES - SCI(t)	65.6	-65.6

Simulated EAC accuracy		
method (PF)	MAPE [%]	MPE [%]
1	1.9	1.0
CPI	2.2	1.7
SPI	16.4	16.4
SPI(t)	14.6	14.6
SCI	16.5	16.5
SCI(t)	14.9	14.9
0.8 CPI + 0.2 SPI	10.7	10.6
0.8 CPI + 0.2 SPI(t)	6.9	6.9

According to the MAPE values[1] the best performance for time forecasting can be expected from the unweighted Earned Schedule method. For cost forecasting the unweighted and CPI-weighted methods should yield the best results.

2.3.2. Tracking description

Tracking authenticity

Manual tracking was performed over 3 tracking periods with a length of approximately 1 month. The Real Duration and Real Cost mentioned in section "2.3.3. Earned Value Management" are based on manual user input.

The tracking information obtained from the project owner and introduced in ProTrack includes actual activity start dates, durations and costs.

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[1] The MAPE gives the best indication for the forecast accuracy (the lower the MAPE, the more accurate the method) since all deviations from the targeted real duration (real cost) are cumulated, whereas for the MPE underestimates can be compensated by overestimates and vice versa, possibly leading to an overly positive evaluation of a certain method. However, the MPE can provide useful information about the nature of the deviations, i.e. does the method rather underestimate or overestimate the real duration (real cost)?

2.3.3. Earned Value Management

2.3.3.1. Performance metrics

	CV [€]	SV [€]	SV(t) [d]	CPI [-]	SPI [-]	SPI(t) [-]	p-factor [-]
avg	44111.5	-16338.5	-45	2.1	0.8	0.8	0.8
std dev	30054.6	19263.3	39.5	0.2	0.3	0.2	0.2
final	74160.0	0	0	2.2	1.0	1.0	1.0

2.3.3.2. Time forecasting

PD	184 days	Real Duration	184 days	Late	0%

EAC(t)			Real Accuracy	
method - PF	avg [d]	std dev [d]	MAPE [%]	MPE [%]
PV - 1	238.0	1.0	9.5	-9.5
PV - SPI	238.7	0.6	9.7	-9.7
PV - SCI	239.0	0.0	9.7	-9.7
ED - 1	239.3	0.6	9.8	-9.8
ED - SPI	240.0	1.0	9.9	-9.9
ED - SCI	241.0	1.0	10.1	-10.1
ES - 1	242.0	1.0	10.3	-10.3
ES - SPI(t)	243.0	1.0	10.5	-10.5
ES - SCI(t)	243.7	0.6	10.6	-10.6

2.3.3.3. Cost forecasting

BAC	132.570 €	Real Cost	58.410 €	Under Budget	-55.9%

EAC			Real Accuracy	
method (PF)	avg [€]	std dev [€]	MAPE [%]	MPE [%]
1	88458.5	30054.6	17.1	-17.1
CPI	64507.8	5464.1	3.5	-3.5
SPI	136826.8	106885.3	44.8	-44.8
SPI(t)	113429.0	63506.0	31.4	-31.4
SCI	88679.4	41542.1	17.3	-17.3
SCI(t)	77042.4	20068.8	10.6	-10.6
0.8 CPI + 0.2 SPI	68602.2	9095.4	5.8	-5.8
0.8 CPI + 0.2 SPI(t)	68233.1	8617.7	5.6	-5.6